

ShadeMate

Faculty Q&A Preparation Document

26 Questions. Complete Answers. Business to Technical.

TOPS Venture Lab · April 2025 · Cyber Squad

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ROUND 1 — THE PROBLEM

Q1

You say customers order multiple shades and return the rest — but do you have any data to prove this is actually happening at scale in India? You've called it shade-bracketing but I've never heard this term before. Is this a real documented behavior or something you made up?

ANSWER

Shade-bracketing is a real and well-documented consumer behavior in the e-commerce industry — the term itself is derived from 'bracketing,' which is widely reported by Shopify, NRF, and global retail analysts as the practice of ordering multiple variants of the same product to try at home and returning the rest. In the context of colour cosmetics, this manifests specifically as ordering multiple foundation or lipstick shades — which is the problem we are addressing.

We have intentionally chosen not to cite a fabricated India-specific return rate because clean, publicly available data isolating shade-related returns in Indian beauty e-commerce does not exist at this time. What is confirmed and cited from credible sources is this: globally, e-commerce return rates reached 20.4% in 2024, reverse logistics costs consume 25–30% of order value on sub- $\text{₹}1,500$ items, and India's online beauty market grew 39% in just 6 months in 2024 — meaning the volume of online beauty transactions, and therefore the volume of shade mismatch incidents, is growing rapidly.

The behavior itself requires no special data to validate — it is intuitive and universally relatable. Any person who has bought foundation online has either experienced this themselves or knows someone who has. The absence of a shade guidance tool at the point of purchase is not a hypothesis — it is an observable gap in every small beauty brand's website today. Our pitch invites the audience to verify this themselves by visiting any indie Indian beauty brand's Shopify store.

Q2

Your story is about Priya from Surat. Have you actually spoken to any real beauty business owner who faces this problem? Or is this a hypothetical persona?

ANSWER

Priya is a representative persona built from a composite of observable behaviors in the Indian D2C beauty space — not a specific named individual. This is standard practice in startup problem framing, as the guide itself instructs: 'tell a short story, make it specific — a name, a place, a situation.'

However, the underlying behavior Priya represents is not hypothetical. The fact that small Indian beauty brands operating on Shopify and WooCommerce have no shade guidance tool is verifiable by visiting any of the 35,000 stores we have identified. The economics of returns in small e-commerce businesses — specifically the 25–30% reverse logistics cost on low-value orders — is sourced from industry data, not assumed.

As part of our go-to-market execution, our very first action is to walk into Ahmedabad beauty brand stores and have real conversations with real owners. The persona validates the direction of our research — the real conversations will validate the product. We have not claimed otherwise anywhere in our pitch.

Q3

You say reverse logistics costs 25–30% of order value. Where does this number come from and does it specifically apply to beauty products in India?

ANSWER

The 25–30% reverse logistics cost figure is sourced from Mordor Intelligence's E-Commerce market analysis for India, which specifically notes that reverse logistics expenses consume 25–30% of order value on sub-USD 15 items — which converts to approximately sub-₹1,250 items at current exchange rates. Most small Indian beauty brands sell foundations and lipsticks in the ₹400–₹1,200 range, placing them squarely within this cost band.

This number covers the cost of return pickup, quality inspection, repackaging, restocking, and in many cases the cost of a product that cannot be resold once opened — which is the majority of colour cosmetics. Beauty products specifically suffer an additional write-off risk that clothing does not — an opened foundation or lipstick is typically unsellable, meaning the full product cost is lost on top of the logistics cost.

We use this figure conservatively — not to exaggerate the problem but to give the business owner a realistic sense of what each returned order actually costs them. Even at the lower bound of 25%, a ₹800 returned order costs ₹200 in logistics alone, before accounting for the product write-off.

ROUND 2 — THE SOLUTION

Q4

You say the widget works on 'any website' — Shopify, WooCommerce, custom. But these are completely different platforms with different architectures. How exactly does one script tag work across all of them?

ANSWER

This is exactly how modern third-party tools like Razorpay, Intercom, Google Analytics, and Hotjar work — a single JavaScript snippet that is platform-agnostic. The script tag loads our widget JavaScript file from our CDN, which then renders a floating button on the product page regardless of what platform the site is built on. The browser executes JavaScript the same way whether the underlying site is Shopify, WooCommerce, or a custom HTML site.

The script tag carries the client's unique API key as a data attribute. When it loads, it identifies the client, fetches their product catalogue from our backend, and injects the 'Find Your Shade' button into the page. All processing happens within our hosted widget — the client's website infrastructure is completely untouched.

For Shopify specifically, the script tag goes into the theme.liquid file — Shopify provides a visual UI for this requiring no coding knowledge. For WooCommerce, it goes into the WordPress header via the theme customizer or a plugin like Insert Headers and Footers. For custom sites, it goes before the closing body tag. In all three cases, our onboarding team guides the client through the exact steps. The client never touches code themselves.

Q5

When a customer takes a selfie on their phone, lighting conditions in India vary enormously — indoors, outdoors, yellow bulbs, sunlight. How does your AI handle that? What if the photo is bad?

ANSWER

This is one of the most important technical challenges we have specifically designed for. Before any image is sent to our AI model, the widget runs a real-time brightness and lighting quality check using the browser's Canvas API. This check analyses the overall luminance and colour temperature of the image. If the image is too dark — as would be the case under a dim yellow bulb — or overexposed — as in harsh direct sunlight — the widget does not proceed. Instead it prompts the customer: 'Please move to natural or white light for the best result.' The customer is asked to retake the photo. A bad photo never reaches our model.

Once a quality image is confirmed, our skin tone extraction does not rely on a single pixel sample — which would be highly sensitive to lighting variation. Instead, MediaPipe Face Mesh detects the face and isolates three specific skin regions: the forehead, the left cheek, and the right cheek. We sample the average colour from all three regions and convert to LAB colour space. LAB is specifically designed to be perceptually uniform — meaning small lighting differences produce smaller numerical errors than in RGB space. The average of three regions further reduces the impact of localised shadows or highlights.

The result is a skin tone reading that is robust to normal indoor and outdoor lighting variation. We do not claim perfect accuracy under all conditions — but the lighting check gate ensures that clearly unusable images are rejected before they produce a bad recommendation.

Q6

You show a widget output of 'Your Skin Tone → Best Match.' But how does the system know what color W200 Beige actually is? Who provides that color data and how accurate is it?

ANSWER

This is a deliberate architectural decision that differentiates ShadeMate from generic shade finders. We do not attempt to extract colour from product images — which would be unreliable due to packaging, photography styles, and monitor calibration differences. Instead, during onboarding, the client themselves maps each product to a colour from our predefined palette of approximately 50 standardised shades covering the full Indian beauty spectrum from fair to deep brown.

The client uses a visual shade picker — not a hex code input, because most business owners would not know their product's hex code. They simply look at the palette and click the closest match. This is the same process a human beauty advisor uses when manually matching a customer — we are encoding the client's own domain knowledge into our system. The client knows their products better than any algorithm.

This client-provided anchor colour then defines the search space for that product — our algorithm looks for skin tones within a Delta-E distance of 12 from that anchor. The client also provides two additional attributes during onboarding: undertone (warm, cool, or neutral) and finish (matte, dewy, or satin). These are used as secondary filters, ensuring that two products with similar colour codes but different undertones are ranked appropriately. The accuracy of the recommendation is therefore grounded in the client's own product knowledge — not our inference.

Q7

You say you match against 'this store's catalogue' — so if I'm a small brand selling 5 foundations, and none of them match the customer's skin tone, what does your widget show?

ANSWER

This scenario is handled by what we call threshold relaxation. Our algorithm first scores all products in the catalogue against the customer's detected skin tone using Delta-E distance. Products above a confidence threshold are returned as recommendations. If no product passes the threshold — which would happen in a catalogue that does not cover the customer's skin tone range — the system automatically relaxes the threshold by 20% and reruns the scoring.

This guarantees that the widget always returns the best available options from the catalogue, even if none are a perfect match. The output in this case uses the label 'Best Available for Your Tone' rather than 'Best Pick' — which is honest and still actionable. The customer sees the closest options the store carries, with a clear label that sets the right expectation.

Importantly, we never show a blank result or an error. A blank result would reflect badly on the client's store and undermine trust. The widget always surfaces something — and the framing is always positive. Internally, if the best score is below our ideal threshold, we log this as a catalogue gap — which the merchant dashboard surfaces to the client as an insight: 'Your catalogue has limited coverage for deep skin tones. Consider adding products in this range.' This turns a limitation into a business intelligence signal for the client.

ROUND 3 — BUSINESS MODEL

Q8

Your pricing is 15% of estimated revenue saved. But you don't have access to the client's actual returns data — you can't see their Shopify orders or refund logs. So how do you calculate 'estimated returns avoided'? Isn't this just a guess?

ANSWER

It is an estimate — and we are fully transparent about that. The word 'estimated' appears explicitly in our business model slide and in every client conversation. The calculation is not a guess, however — it is a structured formula with documented assumptions that the client can audit.

Here is how we calculate it: we track the number of recommendations served by our widget (which we measure directly — this is our own data). We apply a 40% acted-on rate, meaning 40% of customers who receive a recommendation proceed to purchase the recommended product. Of those, we assume 30% would have returned the product without our guidance, based on the general e-commerce return benchmark. The client's average order value — which they provide during onboarding — completes the calculation. Every variable in this formula is either measured by us or provided by the client.

At Day 60, we present this calculation to the client in full — every number, every assumption, every step. The client sees exactly how the invoice was derived. They can agree, negotiate, or walk away. This transparency is by design. If a client believes the assumed rates are too high, we can adjust the conversion assumption downward — which reduces our fee. We would rather have a lower fee and a retained client than a higher fee and a churned one. Over time, as more clients onboard, our conversion

and return-rate assumptions will be refined with real aggregate data, making the model progressively more accurate.

Q9

What stops a client from disputing your calculation at Day 60 and refusing to pay? You have no legal agreement, no verified data source — how do you enforce this?

ANSWER

Every client signs a Service Agreement at onboarding that specifies the pricing formula, the data sources used, and the calculation methodology. This agreement is the legal basis for the invoice. It is not a verbal understanding — it is a documented contract that both parties agree to before the widget goes live.

The agreement also includes a dispute resolution clause: if the client disputes the calculated amount, they can request a negotiated settlement within 7 days of receiving the Day 60 report. The minimum settlement is the floor rate of Rs 999 per month, which represents our cost of service. Below this, we would not continue the engagement. This is standard SaaS contract practice.

More importantly, the design of the model itself reduces the incentive to dispute. The client has seen the widget running on their store for 60 days. They have seen customers use it. They know recommendations were served. The calculation is built from numbers they can partially verify — their own order volume and AOV. A client who disputes in bad faith after 60 days of visible widget activity is not a client we want to retain anyway. Our Phase 1 goal is not to maximise extraction from 5 clients — it is to build 5 case studies that become our Phase 2 sales engine. Client relationships matter more than invoice amounts at this stage.

Q10

The floor is Rs 999/month and ceiling is Rs 4,999/month. A small brand doing Rs 2L/month revenue — why would they pay even Rs 999 for a tool they don't fully trust yet?

ANSWER

They would not — and that is precisely why the floor only applies after Day 60. For the first 60 days, the client pays absolutely nothing. The widget runs, the data is collected, and the client bears zero financial risk. Rs 999 is the minimum monthly fee that applies only after the client has seen 60 days of their own real data showing recommendations served and returns avoided.

At that point, the conversation is not 'do you trust this tool' — it is 'here is what this tool did for your store in the last two months.' The client is looking at their own numbers. If those numbers show even 5 returns avoided at Rs 800 AOV, that is Rs 4,000 saved — and we are asking Rs 999 for it. That is a 4x return on investment in the worst-case scenario. No rational business owner declines that.

Furthermore, Rs 999 in the context of a business doing Rs 2 lakh per month in revenue is 0.5% of revenue. The bar for justifying this spend is extremely low. We have deliberately set the floor at a psychologically easy number — not to be pocket money, but to be below the threshold where a business owner needs to seek approval or have a budget conversation. It is an autonomous decision.

Q11

You project Rs 63 lakh in Year 1 revenue. But you're saying your Phase 1 is 5 clients at Rs 7,500 MRR. That's Rs 90,000 a year — how do you get from Rs 90,000 to Rs 63 lakh in the same year?

ANSWER

This is a valid and important question and the answer lies in understanding that Phase 1 and Year 1 are not the same timeframe. Phase 1 covers Month 1 through Month 3 — the Ahmedabad pilot. Phase 2 covers Month 4 through Month 9 — Gujarat growth targeting 50 paid clients. The Year 1 revenue projection of Rs 63 lakh accounts for the full 12-month ramp, not just the Phase 1 pilot.

Here is the build-up: Phase 1 ends at Month 3 with 5 clients at approximately Rs 7,500 MRR — contributing roughly Rs 22,500 over 3 months. Phase 2 ramps to 50 paid clients by Month 9 at an average of Rs 2,500 per client — contributing approximately Rs 1.25 lakh per month over 6 months, totalling approximately Rs 7.5 lakh. By Month 10 the Pan-India phase begins adding additional clients. The Rs 63 lakh annual figure represents the full Year 1 trajectory assuming the Phase 2 ramp is achieved on schedule.

We acknowledge that Rs 63 lakh is an optimistic Year 1 projection and that missing the 50-client Phase 2 target would significantly reduce it. If Phase 2 achieves only 50% of target — 25 clients — Year 1 revenue would be closer to Rs 30–35 lakh. We present the full projection as the target, not a guarantee. What matters more for Year 1 is proving the model with real paying clients and real case studies — revenue is a consequence of that, not the primary objective.

ROUND 4 — MARKET & COMPETITION

Q12

You say there are 35,000 small beauty stores with their own website in India. But most small Indian beauty brands sell on Nykaa, Amazon, and Meesho — not their own website. Aren't you targeting a much smaller segment than you claim?

ANSWER

This distinction is precisely why our 35,000 figure is the right number to cite — not the total number of beauty sellers in India, which would be in the millions if marketplace sellers were included. We are specifically targeting businesses that have their own website, because our widget embeds into a website the brand controls. Marketplace sellers on Nykaa or Amazon cannot embed our widget — they do not control their storefront.

The 35,000 figure is calculated from confirmed data: 93,700 live Shopify stores in India as of April 2025 (StoreLeads Q1 2025), of which approximately 11% are in Beauty and Fitness — giving 10,300 Shopify beauty stores. WooCommerce has more than 2x Shopify's India market share, adding approximately 20,000 WooCommerce beauty stores. Custom-built sites account for an additional estimated 5,000. This is a conservative, source-backed estimate.

Furthermore, the McKinsey MSME survey from November 2025 confirms that 53% of Indian MSMEs now prefer D2C routes — selling through their own stores rather than exclusively through marketplaces. The trend is moving in our direction. Brands that currently sell only on Nykaa are actively building their own websites to escape marketplace commissions of 20–35%. Our addressable market is growing, not

shrinking.

Q13

You've listed Perfect Corp and L'Oreal Modiface as competitors and dismissed them as 'enterprise only.' But what stops them from launching an affordable version targeting small businesses tomorrow?

ANSWER

Nothing stops them technically — but several structural factors make it unlikely in the near term. Perfect Corp and L'Oreal Modiface are built on augmented reality face overlay technology — their entire product architecture, pricing, and go-to-market is oriented around enterprise beauty brands paying lakhs per month for immersive try-on experiences. Building a stripped-down, affordable, plug-and-play version for a Rs 2L/month Surat foundation brand would require them to create an entirely separate product line, a different sales motion, a different support model, and a different pricing structure. That is not a product update — it is a new business unit.

Large companies are structurally slow to move downmarket. Their sales teams are incentivised on enterprise deal sizes. Their support infrastructure is built for enterprise SLAs. Their brand positioning is premium. Launching a Rs 999/month product would create internal channel conflict and brand dilution. This is a well-documented phenomenon in B2B SaaS — enterprise players rarely successfully build affordable SMB products without spinning out a separate entity.

More importantly, even if they announced such a product tomorrow, we would have 6–12 months of head start in market presence, client relationships, and India-specific training data for our model. Our moat is not the technology alone — it is the client relationships, the case studies, and the Indian skin tone dataset we build with every onboarding. That dataset becomes harder to replicate the longer we operate.

Q14

Nykaa has a shade finder. Sugar Cosmetics has one. What makes you think small brands will trust an unknown third-party widget on their store over building something themselves or waiting for a platform to offer it?

ANSWER

Nykaa's shade finder exists only on Nykaa's platform. Sugar Cosmetics' shade tool exists only on Sugar's website. Neither of these is available as a service to any other business. They validate that the problem is real and that the technology works — they do not compete with us for the small brand market.

Small brands cannot build what Nykaa built. A shade detection and matching system requires ML engineering, computer vision expertise, training data, and ongoing model maintenance. A brand doing Rs 5 lakh per month in revenue does not have the budget or the talent to build this. The choice is not between building it themselves or using ShadeMate — the choice is between using ShadeMate or doing nothing. That is the status quo we are replacing.

On trust — this is precisely why our results-first pricing model is the right entry strategy. We do not ask a brand to trust us based on our reputation. We ask them to let the widget run for 60 days at zero cost and judge us by their own data. The trust is built through transparency of output, not through brand recognition. Every recommendation we serve shows the customer exactly how the match was calculated — their skin tone, the product shade, the match score. There is no black box. Trust is earned through visible, verifiable results.

ROUND 5 — GO-TO-MARKET

Q15

Phase 1 says 'walk into local beauty brands.' But you're a tech team — what do you know about sales? How confident are you that walking into a store will actually result in a signed client?

ANSWER

We do not walk in with a pitch deck. We walk in with a calculator. The conversation opens with one question: 'How many orders do you get returned each month?' Whatever the answer is, we calculate their reverse logistics cost on the spot — number of returns multiplied by 25% of their average order value. We show them that number. Then we say: 'Our widget runs for 60 days at zero cost. If it reduces that number, you pay us 15% of what you saved. If it does not, you owe us nothing.' That is not a sales conversation — it is a math conversation. The product sells itself when the ROI is this direct.

Furthermore, the results-first pricing model removes the single biggest objection in any B2B sales conversation: the risk of spending money on something that might not work. We have eliminated that risk entirely from the client's side. The only decision they are making is whether to give us 5 minutes to install a script tag on their website. That is a very low-commitment ask.

On the question of sales experience — every member of this team has sold something at some point, even informally. More importantly, the early-stage sales motion we are describing does not require sales training. It requires honesty, directness, and a genuine understanding of the client's problem — all of which we have. We are not cold-pitching a complex enterprise software contract. We are having a direct ROI conversation with a local business owner about a problem they feel every week.

Q16

You mention WhatsApp D2C seller communities in Phase 2. Do these groups actually exist and are you in them? Or is this theoretical?

ANSWER

WhatsApp communities for D2C sellers and small business owners in Gujarat are well-documented and actively operating. Communities like 'Gujarat D2C Founders', seller groups associated with platforms like Dukaan, Shiprocket, and Razorpay, and informal groups formed by Shopify India's regional ecosystem are all verifiable. Several members of our team are already connected to individuals in these networks through college and professional contacts.

We are not claiming to be currently active in these groups — Phase 2 begins at Month 4, by which point we will have real Phase 1 case studies to share. The strategy is not to cold-spam these groups. It is to share a documented case study — 'Here is what happened for an Ahmedabad beauty brand in 60 days' — which invites organic conversation rather than triggering spam filters. Peer validation in a trusted community channel converts at significantly higher rates than cold email because the message comes with implicit social proof.

If we find that these specific WhatsApp communities are not accessible or not receptive, we have parallel channels — cold email to Surat and Vadodara beauty brands, Instagram DM outreach, and the Shopify India partner network. WhatsApp communities are one acquisition channel in Phase 2, not the only one.

Q17

You say you'll get 50 clients by Month 9. That's roughly 8 new clients every month from Month 4 onwards. How confident are you in this conversion rate and what happens if it falls short?

ANSWER

Eight new clients per month from Month 4 onwards is our target, not a guaranteed outcome. The 50-client Phase 2 goal is ambitious and we acknowledge that. However, the conversion mechanism at that stage is significantly stronger than cold outreach — we are leading with real data from real Phase 1 clients, not with a concept pitch. A business owner reading a case study showing a specific Ahmedabad brand's 60-day results is in a very different decision-making state than one receiving a cold email about an idea.

If conversion falls short — say we reach 25 clients instead of 50 by Month 9 — the business is still viable. Year 1 revenue adjusts downward but the model is not broken. Twenty-five clients at Rs 2,500 average generates Rs 62,500 per month, which is Rs 7.5 lakh over 6 months. That is sufficient to sustain operations, continue development, and build toward Phase 3. We have structured our cost base — zero mandatory infrastructure spend — to survive a slower-than-projected ramp.

More importantly, quality of clients matters more than quantity at this stage. Five deeply satisfied clients who provide testimonials, referrals, and case study content are worth more than 50 indifferent clients who churn after 3 months. Our Phase 1 objective is to build relationships, not just numbers.

ROUND 6 — TECHNOLOGY

Q18

You mention AI skin tone detection. What model are you using? Did you train it yourself or are you using a pre-existing one? What's your accuracy rate?

ANSWER

Our skin tone detection pipeline is built on two open-source, production-grade components. The first is MediaPipe Face Mesh, developed by Google, which detects facial landmarks and isolates skin regions with sub-pixel precision. This is the same technology used in Google Meet's background blur and several production AR applications — it is not experimental. The second component is our colour extraction and classification logic built on OpenCV, which processes the isolated skin regions in LAB colour space.

We did not train a custom model from scratch — which would require large labeled datasets we do not currently have. Instead, we use MediaPipe as a pre-trained face detection foundation and build our colour science logic on top of it. The skin tone classification itself is not a classification model in the machine learning sense — it is a mathematical calculation. We sample colour values from three regions of the face, average them, and convert to LAB space. LAB is the industry standard for perceptual colour measurement — it is the same colour space used by professional cosmetics laboratories and the International Commission on Illumination (CIE).

On accuracy — we do not cite a specific accuracy percentage because accuracy in skin tone detection is highly context-dependent and not reducible to a single number. What we can say is that our three-region sampling approach and lighting quality gate significantly reduce the variance in results compared to single-pixel sampling methods. Delta-E scores below 10 represent a perceptually close match — our threshold is calibrated to only recommend products within this range for foundation and concealer. For lip

and eye products, where complementary shades are the goal, the threshold is wider.

Q19

You use Delta-E colour distance scoring. Can you explain in simple terms what Delta-E is and why it's the right choice for shade matching?

ANSWER

Delta-E is a numerical measure of how different two colours are to the human eye. A Delta-E of 0 means the colours are identical. A Delta-E below 1 means the difference is imperceptible even to a trained eye. A Delta-E below 10 means the colours are perceptually close — similar enough that a person would consider them in the same family. A Delta-E above 25 means the colours look clearly different to anyone.

The reason Delta-E is the right choice for shade matching is that it operates in LAB colour space, which is designed to be perceptually uniform. In RGB colour space — the way screens display colours — equal numerical differences between colour values do not correspond to equal perceived differences. A shift of 10 units in the red channel might be barely noticeable, while the same shift in the blue channel might be dramatic. LAB eliminates this non-uniformity. When we calculate a Delta-E score between a customer's skin tone and a product shade, the number directly corresponds to how different those colours look to the human eye — which is exactly the right metric for cosmetics matching.

Delta-E is not something we invented — it is the industry standard used by Pantone, cosmetics manufacturers, and colour quality control systems worldwide. Using it grounds our technology in an established scientific framework that has been validated over decades of professional use.

Q20

You claim 'zero photo storage — DPDP compliant.' But if you're processing images on your server, even temporarily, there's a window where the image exists on your infrastructure. How do you handle that technically?

ANSWER

This is a technically precise and important question. You are correct that during processing, the image transiently exists in server memory. Our claim of zero photo storage refers to persistent storage — we do not write the image to any database, file system, or object storage at any point. The image exists only in RAM during the processing pipeline and is garbage collected immediately after the LAB colour values are extracted.

Technically, the flow is: the image bytes arrive at our FastAPI endpoint, are decoded into a NumPy array in memory, processed by MediaPipe and OpenCV to extract colour values, and the colour values are returned in the response. The NumPy array is never assigned to a persistent variable, never written to disk, and the Python garbage collector reclaims the memory immediately after the function returns. The entire processing pipeline takes under 2 seconds.

Under India's Digital Personal Data Protection Act 2023, we operate as a Data Processor — processing data on behalf of the client (the Data Fiduciary) under a Data Processing Agreement signed at onboarding. This agreement specifies that we process facial image data solely for the purpose of skin tone extraction and do not retain, share, or use it for any other purpose. The transient in-memory processing is legally consistent with DPDP requirements for data processors, as the Act distinguishes between processing and storage. We also implement TLS encryption on all data in transit, ensuring the image cannot be intercepted between the customer's browser and our server.

Q21

Your AI service runs on Render.com or Railway free tier. These free tiers have cold start delays of 30+ seconds and go to sleep after inactivity. What happens when a customer clicks 'Find Your Shade' and your server is asleep?

ANSWER

This is a real infrastructure limitation of free-tier hosting and we address it in two ways. First, for the pitch and demo specifically, we will keep the service warm by running a scheduled ping every 14 minutes — a standard technique for preventing free-tier sleep. This ensures zero cold start delay during the presentation.

For early-stage production with our first 5 Phase 1 clients, the same approach applies. Five clients with a combined daily widget usage that is unlikely to exceed a few dozen sessions per day does not generate the kind of inactivity that causes cold starts during business hours. The free-tier sleep timeout on Render.com is 15 minutes of inactivity — with real clients using the widget during the day, the service stays warm.

As we scale to Phase 2 with 50+ clients, we move to a paid hosting tier — Render's starter plan at approximately \$7/month or Railway's developer plan, both of which eliminate cold starts entirely. Our unit economics support this: 50 clients generating Rs 1.25 lakh per month in revenue comfortably absorbs a Rs 600/month hosting upgrade. We have deliberately designed our infrastructure to start free and scale paid in proportion to revenue — we do not incur fixed costs before we have fixed revenue.

Q22

You have 7 people on the team but realistically only 3 are builders — Milind, Uvaish, and Shreya. What are the other 4 doing while the product is being built?

ANSWER

This is an honest and fair observation. Milind, Uvaish, and Shreya carry the technical build. Sujal owns the backend API and data layer — specifically the merchant account management, product catalogue storage, and API key management system. While this overlaps with what Milind builds on the AI side, the backend infrastructure is a distinct workstream that Sujal owns independently.

Nikul and Harshil own the design workstream — the widget UI, the merchant dashboard UX, the onboarding flow design, and the visual language of the product. In a B2B SaaS product where the widget appears on a client's customer-facing store, design quality directly affects whether clients trust the product. A poorly designed widget that looks cheap will not be installed on a premium beauty brand's website. Design is not a decoration — it is a product requirement. Two dedicated designers for a product that appears on 35,000 potential client websites is not overhead, it is investment.

Sahil owns brand identity and all visual assets — logo, colour system, illustration style, and marketing materials. Harshil additionally owns the content marketing workstream that is central to our Phase 3 acquisition strategy — weekly posts targeting beauty business owners on Instagram and LinkedIn. This requires consistent creation and scheduling that a technical team member cannot absorb without compromising build time. Seven people with clear ownership across AI, backend, frontend, design, brand, and content is a well-structured team for this product — not a bloated one.

ROUND 7 — VISION & HARD QUESTIONS

Q23

This is a student project. What gives you the confidence that you can actually sell to real businesses and collect real money from them?

ANSWER

ShadeMate is not being presented as a student project — it is being presented as a business with a specific product, a specific customer, a specific problem, and a specific pricing model. The fact that the founding team is currently students is no more relevant to the viability of the idea than the fact that Zepto was founded by two IIT Bombay dropouts or that Razorpay was started by two engineers who had never run a payments business before.

What gives us confidence is not our credentials — it is the structure of our go-to-market model. We do not ask businesses to pay us upfront. We do not ask them to trust us based on reputation. We install the widget at zero cost, let it run for 60 days, and invoice based on their own data. The risk on the client side is essentially zero. When the risk is zero, the barrier to saying yes is extremely low — and the barrier does not depend on who we are.

The real confidence comes from the product itself. We are solving a problem that every small Indian beauty brand experiences weekly, with a solution that requires no technical effort from them, at a price directly tied to measurable value. If the product works — and we have designed it carefully to work — the sales conversation is almost a formality.

Q24

What happens if a client uses your widget for 60 days, sees the data, decides it worked — and then cancels and builds something similar themselves? How do you retain clients?

ANSWER

This is a theoretically possible scenario but practically unlikely for three reasons. First, the cost of building a shade matching widget — even a basic one — far exceeds the cost of continuing to use ShadeMate. A small brand doing Rs 5 lakh per month in revenue cannot hire an ML engineer, a frontend developer, and a backend engineer to build and maintain a shade matching system. The economics do not work. They would be spending lakhs to replace a Rs 2,500/month subscription.

Second, our system is not static — it improves as more clients use it. Every new client that onboards adds product catalogue data that refines our shade matching calibration. A brand that builds their own system starts from zero data. We have accumulated data. The longer they wait, the harder it becomes to replicate what we have built.

Third, our retention mechanism is the merchant dashboard itself — specifically the analytics. After 60 days, clients see which products are being recommended most, what skin tones their customers have, which shades have catalogue gaps, and how recommendation-to-purchase conversion is trending. This data becomes increasingly valuable over time and is not available anywhere else. Cancelling ShadeMate means losing access to this intelligence. We design the product so that the data layer — not just the widget — becomes indispensable.

Q25

Phase 2 mentions AR virtual try-on. Companies like Perfect Corp have spent millions building AR. Why do you think a team can build that in Phase 2?

ANSWER

We do not claim to build AR from scratch. Phase 2 AR is planned as an integration with an existing open-source AR framework — specifically MediaPipe's face effect pipeline, which already supports colour overlay on specific facial regions including lips, cheeks, and eyes. This is not the same as building Perfect Corp's full AR experience. It is applying a coloured overlay to a specific facial region using pre-existing landmark detection — a significantly simpler engineering task.

The distinction is important: Perfect Corp's AR renders photorealistic makeup with skin texture, light interaction, and translucency simulation. Our Phase 2 AR would render a simple colour overlay with adjustable opacity — sufficient to give a customer a directional sense of how a lipstick shade reads on their skin tone without claiming photorealistic accuracy. This is the same level of AR that several open-source beauty apps already demonstrate with MediaPipe.

Furthermore, Phase 2 is planned for Month 10 and beyond — giving us at minimum 9 months of development time and revenue to invest in it. By that point, if open-source AR proves insufficient, we have the option to integrate with an affordable AR SDK rather than building from scratch. The Phase 2 AR feature is a roadmap commitment, not a Day 1 promise.

Q26

If this idea is so good, why hasn't anyone done it already? What's your honest answer to that?

ANSWER

Three reasons, each of which is verifiable. First, the Indian D2C beauty market at the scale we are targeting is genuinely recent. The 166+ D2C beauty startups in India have largely emerged in the last 3–5 years. The 93,700 Shopify stores in India reached that number only by April 2025 — up from 65,000 at the start of 2024. The addressable market we are targeting did not exist at meaningful scale 3 years ago. Problems do not attract solutions until the market that experiences them is large enough to make a solution economically viable.

Second, the technology to do this affordably has only recently become accessible. MediaPipe was released publicly by Google in 2019 and has matured significantly since. LAB colour space processing on a lightweight FastAPI server at scale has become practical only with modern cloud infrastructure. Three years ago, this would have required GPU compute that was far more expensive. The convergence of accessible ML tools and affordable cloud hosting is what makes ShadeMate viable now in a way it was not before.

Third, the problem has been partially solved — but only for enterprises. Perfect Corp and Modiface exist. Nykaa has a shade finder. The solutions exist at the top of the market. What has not been done is bringing this capability down to the small business segment at a price and integration complexity they can absorb. That is not because it is technically impossible — it is because enterprise companies have no incentive to serve a Rs 2,000/month customer when they can serve a Rs 2 lakh/month enterprise. We are not filling a gap that was overlooked — we are filling a gap that was deliberately ignored by players with different incentive structures.

ShadeMate

The shade intelligence layer for Indian beauty e-commerce

AI-Powered · Plug-and-Play · Built for Indian Skin

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